Dual-Agent-Based Robotic Ultrasound Path Planning and Interaction Control in External-Vision-Independent Environments

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***Abstract*—Ultrasound imaging has emerged as a crucial tool for the diagnosis and navigation of spinal diseases. However, high-quality image acquisition heavily relies on experienced sonographers, which restricts its further popularization. In this paper, a robotic system designed for automated spinal ultrasound scanning is proposed. Drawing inspiration from the spinal anatomy and the actions of seasoned sonographers, the system integrates both a deep learning agent and a reinforcement learning agent to collaboratively guide the adjustment of the ultrasound probe in external-vision-independent environments, relying on real-time ultrasound images and contact force. Then, a hybrid force-to-velocity control framework is proposed to ensure proper ultrasound coupling during the scanning process. Experimental results on a phantom and human participants demonstrated that this system can accurately track spinal features (mean error: less than 1 mm) and maintain normal probe orientation (out-of-plane angular error: 1.61 $\pm$ 1.1°, in-plane angular error: 1.27 $\pm$ 0.9°), resulting in high-quality and reproducible ultrasound images. Overall, our system shows great potential for clinical applications.**

***Note to Practitioners*—This paper is motivated by the increasing needs of human-robot interaction in medical applications, with a specific emphasis on robotic ultrasound imaging. Clinical sonographers suffer from repetitive workload during the diagnostic process, highlighting the significance of automated scanning solutions. In this work, we propose a modular control framework for ultrasound probe positioning that supports a multi-modal autonomous ultrasound scanning system. The system operates independently of external optics or prior geometric knowledge of the scanning object. Comprehensive experimental results demonstrate that the system can effectively cover the region of interest of the spine, facilitating high-quality ultrasound image acquisition and related disease assessment. This advancement is expected to enhance the efficiency of human-robot interaction in healthcare settings and holds promising clinical applications. Furthermore, our research offers valuable insights for the implementation of robotic ultrasound scanning systems applicable to other human tissues.**

***Index Terms*—Interaction control, medical robotics, robot decision-making, robotic ultrasound.**

I. Introduction

SPINAL diseases represented by scoliosis and tumors are prevalent worldwide, affecting more than 900 million people, with roughly 20% of these cases necessitating surgical intervention [1], [2]. While X-ray currently serves as a prevalent navigation technique in minimally invasive spine surgery, it comes with the drawback of exposing patients and clinicians to radiation over prolonged periods [3]. Ultrasound (US) has emerged as a promising surgical navigation tool due to its low cost, radiation-free, and accessibility [4], [5], [6], [7]. However, the quality of US images heavily depends on the operator's skill, and achieving precise control of acquisition parameters remains a challenge, even for seasoned sonographers [8]. To address these hurdles, automatic robotic ultrasound scanning systems (ARUSS) have been introduced in scanning bones, offering superior accuracy and repeatability.

A crucial prerequisite for high-quality image acquisition is to plan the scan path to ensure comprehensive coverage of the region of interest (ROI) [9]. Current path planning approaches often rely on global data from preoperative CT/MRI scans or surface data from external RGB-D cameras. Hennersperger et al. [10] planned the scan path utilizing preoperative MRI data and subsequently transferred it to the robot's task space via surface registration with real-time RGB-D images. Similar approaches have been employed by [11] and [12]. However, in cases such as scoliosis diagnosis, preoperative medical images are usually unavailable. As a result, path planning directly using patient surface data from RGB-D cameras has been developed [13], [14]. Given the inherent variability in individual anatomy and potential body movements, performing scanning tasks based on predefined trajectories is challenging. To address this issue, Zhou et al. [15] proposed a two-step scanning strategy that integrates initial positioning based on RGB-D data with servo control guided by US images to achieve comprehensive thyroid screening. This approach demonstrates the potential of real-time US image information to enhance probe pose control

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in the presence of anatomical variability and body movements. Similarly, Chatelain et al. [16], [17] used a real-time parametric active contour algorithm to dynamically adjust the probe position, thereby ensuring the target remains centered within US images. Although this method is effective, it takes approximately 10 seconds to reduce the centering error to below 3 mm, indicating that there is still room for improvement in terms of time efficiency. Yang et al. [18] extended this approach to spinal scanning tasks using deep learning (DL). However, the chain-like anatomical structure of the spine [19] was not considered, which could result in the loss of target structure during the scanning process. This limitation highlights the potential of leveraging such anatomical priors to enable more reliable scan path planning, particularly in the absence of external vision.

Another key technique for high-quality image acquisition is to maintain the correct orientation of the probe relative to the scanned surface [20]. When the relative angle is close to 0°, high-quality US images can be achieved. This is particularly crucial in orthopedic applications, as the distinct reflection characteristics at soft tissue-bone interfaces [21]. While the human back is often assumed to be flat, it actually presents physiological curvature [22] that necessitates active probe orientation adjustment during scanning. Various strategies have been proposed to optimize probe orientation. Based on surface data from an RGB-D camera, Huang et al. [13] adjusted the probe orientation by estimating the normal vector of the triangular plane around the scanning point. However, the vision-based methods tend to overlook deviations in the normal vector caused by contact deformation. Recent advancements have introduced real-time information, including US images [23], [24], [25] and contact force [26], [27], to guide the probe orientation optimization. Huang et al. [28] categorized US images into diverse suboptimal types by detecting vertebral landmarks, and subsequently adjusted the probe orientation to an optimal state using corresponding strategies. Similarly, Jiang et al. [26] developed a mechanical force model for probe-tissue contact to accurately adjust the probe orientation. However, obtaining a precise estimation of the normal vector necessitates rotating the probe at specific positions, which limits its clinical applicability. In addition, learning-based techniques have shown potential for enhancing stable US scanning in unstructured environments [29], [30], [31]. Ning et al. [27], [32] employed a reinforcement learning (RL) agent to simultaneously adjust probe orientation and contact force. However, this coupled adjustment method may exacerbate the detrimental effects of unstable contact force on orientation adjustment, ultimately restricting the robustness and efficiency of probe orientation optimization.

Due to the attenuation characteristics of US imaging, the quality of US images is significantly affected by the stability of probe-tissue contact [33]. Previous research has primarily focused on maintaining stable contact force during scanning via mechanical structures [22], [34] or external control [35], [36]. External control, renowned for its simplicity, has garnered widespread attention. Gilbertson and Anthony [35] utilized a single degree of freedom force control algorithm on a handheld US device to maintain constant contact force under external interference. Compliance control algorithms [37], [38], such as impedance control and admittance control, have also been introduced to address this problem. Wang et al. [39] demonstrated full-coverage scanning of the breast by the integration of a predefined path from a 3D point cloud with an admittance control algorithm. However, this approach may encounter constraints in situations where the external visual environment is disturbed or inaccessible.

In order to eliminate the aforementioned obstacles, we developed an end-to-end framework for ARUSS based on dual agents, aiming to achieve high-quality spinal US imaging in external-vision-independent environments for diagnostic or surgical tasks. The proposed approach is inspired by the principles of manual sonographic techniques and the intricate structural characteristics of the spine. Although the focus of this work is exclusively on the spine, the framework has the potential for broader application in US imaging of diverse human tissues. The main contributions of this article are listed as follows:

1) To enable effective scan path planning in external-vision-independent environments, a multi-task DL model and an anatomy-aware feature tracking algorithm are devised to leverage the chain-like characteristics of the spine for reliable probe guidance.
2) To accurately establish the desired relative orientation between the US probe and human tissues, an orientation-sensitive RL agent is proposed to simulate the probe-tissue contact dynamics, subsequently establishing the relationship between the applied contact force and the resulting probe orientation.
3) To maintain a normal probe orientation and a stable contact force during US scanning, a hybrid control framework integrating the aforementioned RL agent and a force-to-velocity (F2V) algorithm is introduced to reduce decision-making anomalies in the RL agent arising from fluctuating contact forces.

The rest of this article is organized as follows. Section II describes the system framework. Section III presents the spinal scan path planning algorithm. Section IV introduces the probe orientation adjustment approach via the RL agent. Section V briefly discusses the F2V algorithm and the implementation details of the closed-loop controller. Experimental results are presented in Section VI. Finally, Section VII concludes this article.

II. System Overview and Calibration

The proposed ARUSS consists of four main components: a robotic arm (UR3, Universal Robots, Denmark), a US imaging system (HD60, Hisense Medical, China), a force/torque (F/T) sensor (M4325B, SRI, China) and a desktop computer (Intel i9-13900 CPU, NVIDIA RTX 4060 GPU), as shown in Fig. 1.

The proposed ARUSS integrates force and US image data for precise and repeatable probe pose control in environments without external vision. This integration enhances the clarity of input data for downstream tasks, such as image segmentation and registration, which are crucial for precise navigation in orthopedic procedures. It is worth noting that

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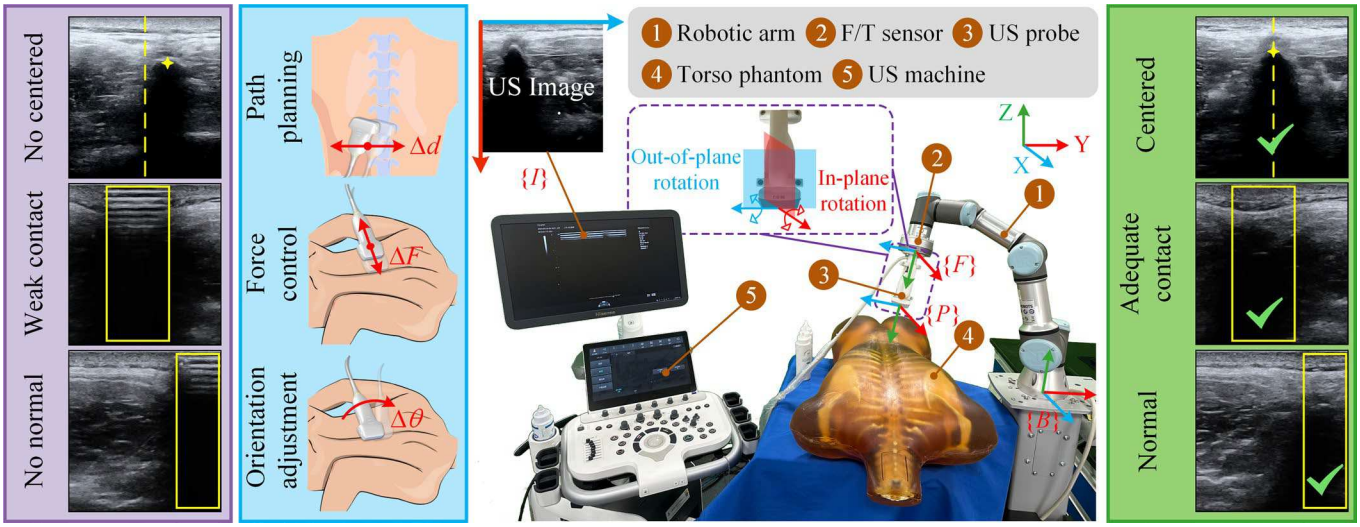


Fig. 1. Overview of the proposed ARUSS, including US image artifacts, corresponding patterns, and adjusted results.

a decoupled framework was adopted to handle the distinct characteristics of US image and force information, rather than a single unified framework. Specifically, a DL agent is employed for probe positioning based on US images, while a RL agent is utilized for probe orientation control based on force feedback, reflecting the distinct nature of these tasks. This separation effectively mitigates the dimensional asymmetry between US images and force data, which complicates joint training. It also enhances the modularity of the system, enabling flexible adaptation to different anatomical regions through selective updating of task-specific components. Furthermore, the unstructured nature of flexible human tissue poses a challenge for achieving accurate force analysis during continuous scanning. To avoid artificial modeling of the scanning process, a discrete single step scanning strategy (4S) is utilized instead of complete trajectory generation. The 4S methodology allows for probe orientation adjustment based on real-time force before proceeding to the subsequent scanning position. Overall, the ARUSS for the spine has three basic functions: ensuring that the spinal feature remains centered in the view during scanning, keeping a normal probe orientation, and maintaining stable contact force between the probe and human tissues for optimal acoustic coupling.

To control the robot using US images, the pixel positions must be mapped from the US image to the robot base frame as:

$$\mathbf{T}_I^B = \mathbf{T}_F^B \cdot \mathbf{T}_P^F \cdot \mathbf{T}_I^P \tag{1}$$

where $\mathbf{T}_i^j$ is the transformation matrix transferred from frame $\{i\}$ to frame $\{j\}$.

III. Scan Path Planning via Deep Learning Agent

To devise an optimal scan path that covers the ROI while simultaneously compensating for any potential target motion, a DL-based online planning technique for optimizing US scan paths is developed in this section, as shown in Fig. 2.

A. Multi-Task Network for Spinal Features

The predefined scan path may be ineffective in scenarios involving unexpected motion, posing particular challenges in conditions such as scoliosis, where accurately predicting spinal position is difficult. For continuous coverage of the ROI, the application of real-time image analysis becomes imperative. In spinal US imaging, the spinous process (SP) can be regarded as a pivotal anatomical feature. The stability of the ARUSS can be greatly improved by ensuring that the SP remains centered within US images. As a result, we investigated a DL agent to extract the position of the SP from each US image.

Inspired by facial keypoint detection, we employed a heatmap-based approach for the SP localization. The heatmap was generated by using the Gaussian distribution centered around the SP label, where each pixel indicated the probability that the corresponding location represented the SP. In contrast to direct CNN-based coordinate regression, the heatmap-based approach encourages the network to focus more intently on the ROI, leading to more precise and stable predictions. In addition, considering the real physical attributes of the spine, spinous processes are not continuously distributed. A single-task keypoint localization network may not provide meaningful position information within intervertebral gaps (IVG). Hence, we defined two distinct views derived from the structural information in US images: the SP view and the IVG view.

To simplify the model, a multi-task network instead of separate networks was proposed for SP localization and view classification. Given that previous research has demonstrated the superior performance of ResNet [40], [41] on US images, the pre-trained ResNet18 was selected as the encoder for extracting shared features. For the regression of the heatmap in the feature localization, the FPN architecture [42] was utilized as the decoder. The overall network structure can achieve a good balance between performance and efficiency. In addition, a two-step training method was proposed to mitigate overfitting with limited samples: pre-training on a mixed dataset including two types of views, followed by

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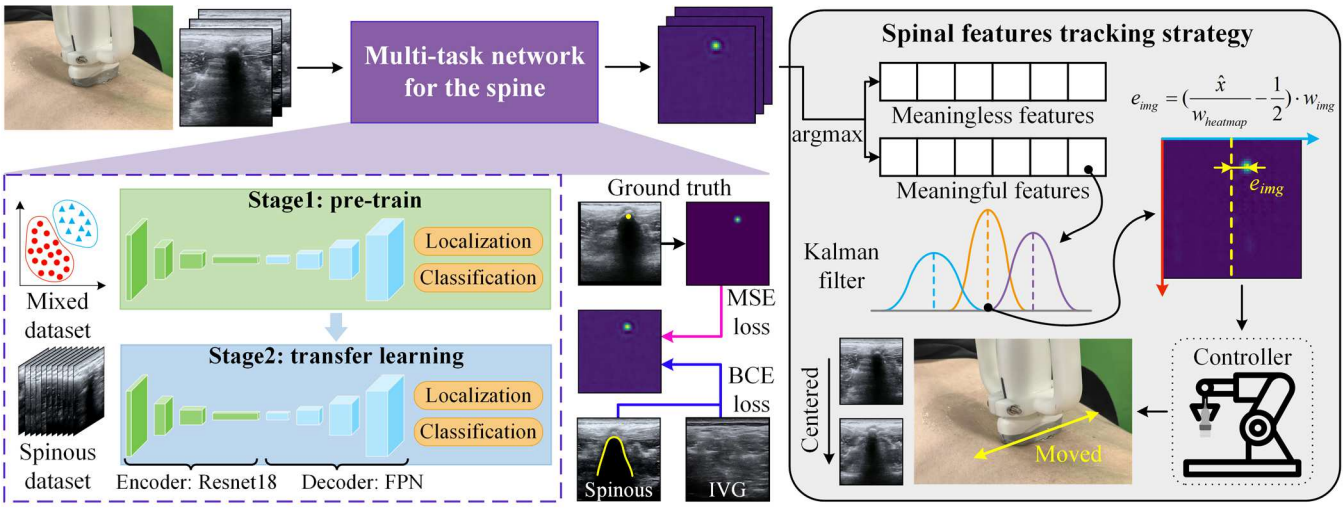


Fig. 2. Workflow of the scan path planning, including the multi-task network and feature tracking algorithm.

fine-tuning on a smaller SP dataset. Using customized coefficients for each stage, the same composition of the loss function was utilized in both training stages as follows:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{loc} + \lambda_2 \mathcal{L}_{cls} \tag{2}$$

where the localization loss function $\mathcal{L}_{loc}$ is the mean square error (MSE) loss, and the classification loss function $\mathcal{L}_{cls}$ is the binary cross-entropy (BCE) loss.

B. Control Strategy for Spinal Features Tracking

The analysis of US images provides guidance for robotic path control. The captured US images serve as inputs to the proposed multi-task network for the generation of real-time predictive heatmaps and view classification. The coordinate of the SP corresponds to the index value of the largest element in the heatmap, which can be calculated as follows:

$$pose(x, y) = \arg\max(heatmap) \tag{3}$$

where x and y represent the horizontal and vertical coordinate positions of the SP in the frame of US image $\{I\}$, respectively.

Due to the existence of IVG, the meaningful information obtained from the DL agent is discrete. To prevent the loss of SP during tracking, a Kalman filter was used to generate a continuous path. When the US image contains SP (i.e., the classification is the SP view), the position of SP derived from the neural network is integrated with the position estimated by the Kalman filter to determine the position of SP at the next moment. Otherwise, the position of SP at the next moment is predicted based on the position estimated by the Kalman filter and the velocity of the robot. Furthermore, the integration of the Kalman filter can also effectively link the outputs of consecutive frames, thereby providing more stable trajectories for the robot.

Full coverage of the spine can be confirmed when the SP is aligned with the centerline in the width direction of US images. Once meaningful location information is obtained, the pixel error between the SP and the centerline can be calculated by:

$$e_{img} = \left(\frac{\hat{x}(k)}{w_{heatmap}} - \frac{1}{2} \right) w_{img} \tag{4}$$

where w_{img} and $w_{heatmap}$ denote the pixel width of the US image (430 pixels) and the pixel width of the predicted heatmap (56 pixels), respectively. $\hat{x}(k)$ is the position of SP along the X-axis in control cycle k. The pixel error e_{img} must be mapped to the error in the physical world Δx using the scale factor s_x before being used for robot. To avoid the problem of robotic trembling arising from high-frequency position control, the position error Δx is converted into a velocity command, thereby facilitating a smoother motion. To further enhance the stability of velocity, a first-order filter was implemented as follows:

$$\Delta x = s_x e_{img} \tag{5}$$

$$v_y^B(k) = \alpha v_y^B(k-1) + (1-\alpha) K_x \Delta x \tag{6}$$

where $v_y^B(k)$ denotes the discrete velocity along the Y-axis in the robot base frame $\{B\}$, α is a filter parameter, and K_x is the proportional control gain between position and velocity. During continuous scanning, the velocity v_x^B along the X-axis in the base frame $\{B\}$ is preset to a constant value.

IV. Normal Orientation Adjustment via RL Agent

Previous study has demonstrated that deviations of the probe exceeding 15° from the normal direction can significantly hinder the visualization of the bone boundaries [26]. Therefore, achieving an optimal probe orientation is crucial for obtaining high-quality US imaging. While previous efforts have explored various approaches [26], [39] to achieve this objective, most rely on explicit mechanical modeling or complex controller parameter tuning. Such reliance limits their generalizability and practicality in the presence of nonlinear tissue contacts. In contrast, the RL-based approach eliminates the need for explicit modeling or parameter tuning by directly learning an optimal orientation adjustment strategy. It holds promise

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for enabling accurate and robust probe orientation control, particularly in unstructured human tissue conditions. This section presents an RL-based algorithm designed for real-time probe orientation adjustment, enabling it to dynamically follow the curvature of human back.

A. Problem Formulation

For US imaging of bone, the probe orientation is typically optimized to maintain a perpendicular alignment with the constraint surface, ensuring optimal imaging quality. In external-vision-independent environments, the sole reliance is placed on contact force information to direct the adjustment of probe orientation. To establish the nonlinear relationship between contact force and relative probe orientation, an RL agent was developed for implicit modeling of the contact environment.

1) States and Observations: The full state of the agent can be characterized by the actual and desired probe orientation, along with the corresponding contact force. However, as the inherent unobservability of the constraint surface, the agent is unable to directly perceive the desired probe orientation and contact force. Therefore, in our system, the real-time contact force $\mathbf{F} \in \mathrm{R}^6$ acquired by the agent was used as the observation at timestamp t, facilitating the formulation of subsequent action decisions [43].

2) Actions and State Transition: The policy $\pi(a_t|s_t)$ is a mapping from the current observation s_t to an action decision a_t. Traditionally, the robot actions within the world frame have been represented by a transformation matrix $\mathbf{T}^P$, which serves as the basis for computing the subsequent probe pose by $\mathbf{T}_P^B \leftarrow \mathbf{T}_P^B \mathbf{T}^P$. Although not all 16 parameters within this matrix are necessary for orientation control, directly learning parameters in the rotation matrix remains challenging. This stems from inherent mathematical constraints of the rotation matrix that may be violated due to numerical instability in predictions, thereby leading to unacceptable orientation changes in clinical settings. To address this issue, embedding prior constraints into the training process could be a promising approach. It simplifies training by reducing the dimensionality of the parameters to be predicted and mitigates the risk of uncontrollable orientation changes. While prior constraints help improve training stability, practical deployment requires additional considerations, such as hardware compatibility. Specifically, on the UR robotic platform, position-based commands can introduce unintended oscillations, potentially compromising task performance. To address these challenges, we deployed a strategy where the action is directly applied to the end-effector's velocity. This approach strikes a balance between practical implementation and training effectiveness, ensuring the feasibility and reliability of the robotic system in real-world applications. Considering safety constraints in real US scanning, the outputs were further set to control torques rather than angular velocity values. Specifically, these torques were converted into angular velocities using a compliance control framework (as detailed in Section V), enabling the probe to rotate around its tip in a controlled manner. Furthermore, the maximum control torque was limited to ensure patient safety. Subsequent to the rotation, new contact force data can be obtained from the sensor and used to update the observations.

3) Reward: The rigidity in the back tissues prompts the generation of contact torques $\{T_{rx}, T_{ry}\}$ upon the application of a constant force. Friction can be ignored because the contact region is usually pre-coated with US gel. Consequently, it is indicated that the desired contact torques should be 0 when the probe orientation is normal to the constraint surface. To achieve this, the reward function devised for training should encourage the agent to minimize contact torques as follows:

$$\mathcal{R}_{torque} = -\left\|(T_{rx}, T_{ry})\right\|_2 \tag{7}$$

where $\|\cdot\|_2$ represents the L2 norm. To facilitate the learning process of the agent, a sparse reward of + 100 is assigned for goal achievement on the dense reward. In addition, penalties were imposed for safety-related constraints [44]. For instance, the loss of probe contact or exceeding the normal contact force threshold would obtain a penalty (-1) for the agent. These reward parameters were initially established according to the work of [29] and were further validated for their effectiveness in guiding the learning of the agent within simulation settings. Hence, the reward at any given timestamp t can be formulated as follows:

$$\mathcal{R}_t = \begin{cases} \mathcal{R}_{torque} - 1, \text{if the force is not desired} \\ \mathcal{R}_{torque} + 100, \text{if the objective is reached} \\ \mathcal{R}_{torque}, \text{otherwise} \end{cases} \tag{8}$$

4) Terminations and Truncations: Three termination criteria were set: 1) the target with the threshold set to 0.02 N·m, empirically determined through extensive experimentation to strike a balance between stability and efficiency, is achieved. 2) the probe exceeds the predefined maximum range of motion. 3) the maximum time step is reached.

B. Deep Reinforcement Learning Algorithm

The RL process was proposed within the framework of the Markov Decision Process [27]. It can be described by $\mathcal{S}$, $\mathcal{A}$, p, $\mathcal{R}$, and γ, where $\mathcal{S} = \{s\&_1, s_2, \ldots, s_t\}$ represents the state space, $\mathcal{A} = \{a_1, a_2, \ldots, a_t\}$ is the action space, $p(s_{t+1} \mid s_t, a_t)$ denotes the state transition probability of a given action, $\mathcal{R}(s, a) = r$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. The objective of the training process was to identify the policy with optimal parameters θ to maximize the expected return, as follows:

$$\theta^* = \arg\max_{\theta} \mathrm{E}\left[\sum_t \gamma^t r_t\left[s_0, \pi_\theta\right]\right] \tag{9}$$

This optimization problem can be addressed by gradient ascent, using the following formula to derive the gradient of the objective function:

$$\begin{aligned} \nabla_\theta J(\theta) &= \sum_{s\in\mathcal{S}} p_{\pi_\theta}(s) \sum_{a\in\mathcal{A}} Q_{\pi_\theta}(s, a) \nabla_\theta \pi_\theta(a|s) \\ &= \mathrm{E}_{\pi_\theta}\left[Q_{\pi_\theta}(s, a) \nabla_\theta \log \pi_\theta(a|s)\right] \end{aligned} \tag{10}$$

where $p_{\pi_\theta}(s) = \sum_t \gamma^t p(s_t = s|s_0, \pi_\theta)$, $Q_{\pi_\theta}(s, a)$ is the state-action value function. Considering the limitations of training

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in real environments, the proximal policy optimization (PPO) method based on actor-critic structure was utilized [45]. The optimization objective in the form of PPO-Clip was as follows:

$$\theta^* = \arg\max_{\theta} \mathrm{E}_{s\sim p_\pi}\mathrm{E}_{a\sim\pi_{\theta old}(\cdot|s)}\Big[\min\Big(\frac{\pi_\theta(a|s)}{\pi_{\theta old}(a|s)}A_{\pi_{\theta old}}, \times \mathrm{clip}\left(\frac{\pi_\theta(a|s)}{\pi_{\theta old}(a|s)}, 1-\varepsilon, 1+\varepsilon\right)A_{\pi_{\theta old}}\Big)\Big] \tag{11}$$

where $A_{\pi_{\theta old}} = A_{\pi_{\theta old}}(s,a)$ denotes the advantage estimator. The policy was set up as a multilayer perceptual model. On this basis, the agent can continuously optimize the policy to autonomously adjust the normal probe orientation.

V. Contact Force and Overall Control Strategy

As another pivotal factor in image quality, stable contact force is crucial for obtaining US images and observations for the aforementioned tasks. To ensure optimal acoustic coupling during continuous scanning, this section introduces the F2V control algorithm and the implementation logic of the ARUSS.

A. Compliance Control for Stable Contact Force

Traditional admittance control involves the modeling of the end-effector as a spring-damper system and adjusting three parameters to modify its dynamic characteristics:

$$F_d = M(\ddot{z} - \ddot{z}_d) + B(\dot{z} - \dot{z}_d) + K(z - z_d) \tag{12}$$

where F_d denotes the desired force. M, B, and K are the virtual inertia, virtual damping, and virtual stiffness. z and z_d denote the current pose and desired pose, respectively. To mitigate oscillations, a velocity-based admittance control approach was proposed. In the acoustic coupling task, where predefined motion trajectories are absent, the robot's movement is adjusted in response to the deviation between the actual and desired contact force. The F2V method discretizes (12) by setting $\ddot{z}_d = \dot{z}_d = z_d = 0$ and $K = 0$ to establish the relationship between contact force error and robotic velocity as follows:

$$\Delta F_z = F_z - F_{z_d} = M \cdot \frac{(v(k) - v(k-1))}{\Delta T} + B \cdot v(k) \tag{13}$$

The velocity along the Z-axis direction for robot control can be calculated as follows:

$$v_z(k) = \frac{\Delta F_z + M \cdot v_z(k-1)/\Delta T}{B + M/\Delta T} = \alpha \Delta F_z + \beta v_z(k-1) \tag{14}$$

In general, the F2V algorithm directly translates force values into velocity commands, driving the robot to adapt to variations in tissue properties. When the end-effector is not in contact with the environment, the robot operates in free-motion mode until the actual contact force matches the desired force. To maintain the stability and precision of force data, all signals are denoised before returning to the controller, and the gravitational effect on the probe is compensated using the method in [26].

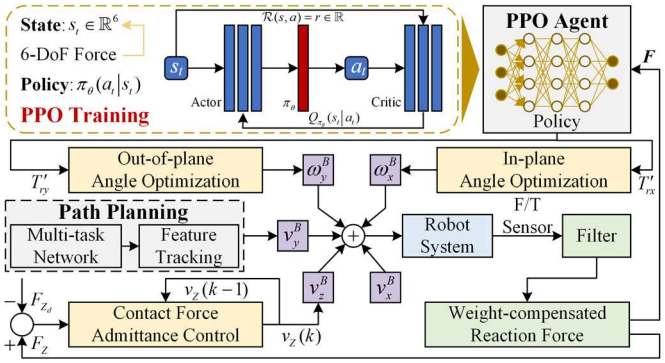


Fig. 3. Control logic of the proposed ARUSS.

B. Overall Control Logic

The commands for path tracking, orientation adjustment, and stable contact force are decoupled from each other. This decoupling allows for the adjustment of the probe orientation, with or without the aid of US images, as the RL agent relies solely on force data for its decision-making. However, to ensure smooth imaging, these commands are combined before being given to the controller. As shown in Fig. 3, the combined linear velocities for position control and angular velocities for orientation control are input into the controller for execution. Moreover, considering patient safety and the primary objective of our system, a safety mechanism is implemented to address potential probe occlusions or hardware malfunctions. In the event of such incidents, the system immediately halts operation and prompts manual inspection before resuming the scanning process. To optimize computational efficiency, the decision-making processes of the dual agents operate concurrently in parallel. This parallel structure results in a total latency of approximately 6.21 ± 4.13 ms between US image acquisition and control command output. Despite this latency, the system maintains high scanning accuracy and stability, as evidenced by the subsequent experimental results in Section VI.

VI. Experiment and Result

The method and algorithm proposed in this study are evaluated with the ARUSS shown in Fig. 1. Datasets used for training the DL agent included 5193 spinal US images obtained from a phantom (CT-A02, Truephantom, Canada) and 3440 spinal US images sourced from 5 young volunteers. Our researchers explained the entire process to all volunteers and obtained their consent. All US images and labels were collected under the guidance of experienced sonographers. The original US images are 430×455 pixels in size, which are normalized and resized to 224×224 pixels before input to the network. The training setup involved using the Adam optimizer with a learning rate of 0.0003 for 30 epochs with batch size 12. To prevent overfitting, 5-fold cross-validation was employed. The entire process was executed on a server (NVIDIA Tesla V100, 32GB). Given challenges in accurately obtaining the flexible environment and contact force in simulation, the RL agent was trained for 35,000 timesteps in a real environment. A total of four sets of experiments

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TABLE I
PERFORMANCE COMPARISON OF VARIOUS NETWORKS

Dataset	Network	Multi	Classification accuracy	PCK	FPS
Phantom	ResNet18	×	99.20%	-	239
	ResNet18-FPN	×	-	98.72%	203
	Our method	√	99.79%	99.54%	186
Human	ResNet18	×	98.53%	-	220
	ResNet18-FPN	×	-	93.74%	198
	Our method	√	98.87%	95.26%	169

were conducted, including scanning path planning, orientation adjustment, contact force control, and assessment of US image quality.

A. Effectiveness of the Scan Path Planning

1) Evaluation of the multi-task network: To evaluate the performance of the proposed multi-task network, a control experiment using single-task networks was conducted. Inspired by the percentage of correct keypoints (PCK), the localization accuracy was determined by measuring the distance between detected and actual SP values and comparing it against a predefined threshold. Real-time performance was quantified in terms of frames per second (FPS). The comparisons are shown in Table I. Across both phantom and human datasets, our method exhibited a higher accuracy in view classification compared to the ResNet18, with gains of 0.59% and 0.34%, respectively. Furthermore, compared to the ResNet18-FPN, our method demonstrated enhanced accuracy in SP localization by 0.82% and 1.52% for the two datasets. The results indicate the efficacy of the proposed multi-task network in outperforming single-task networks, meeting accuracy and real-time usage requirements in practical scenarios.

2) Evaluation of the feature tracking strategy: To validate the effectiveness of the proposed approach in maintaining SP features in the center of the US view, we conducted automatic scanning trials on a phantom and human volunteers with diverse BMI levels. The phantom trials evaluated the robustness of our tracking strategy across three types of scoliosis (mild: <20°, moderate: 20°-40°, severe: >40°). In human trials, volunteer postures were modified to simulate various scoliosis conditions, testing the generalizability of the system across different patient profiles. Experimental results are presented in Fig. 4. The comparison between the detected SP and actual SP positions from US image sequences, as shown in Fig. 4(b), illustrates high localization accuracy of our method in practical scenarios. However, the existence of noisy detection outcomes may have an impact on robotic control and feature tracking. Consequently, to address this problem, the Kalman filter was employed to generate continuous paths, resulting in an effective reduction in noise interference, as shown in Fig. 4(b). This observation is reinforced by the actual robotic motion trajectory illustrated in Fig. 4(a), especially at the markers corresponding to the IVG. Notably, trials involving volunteers with different BMIs demonstrated that our approach maintains high tracking accuracy regardless of individual anatomical variations. During continuous scanning, the average real-time

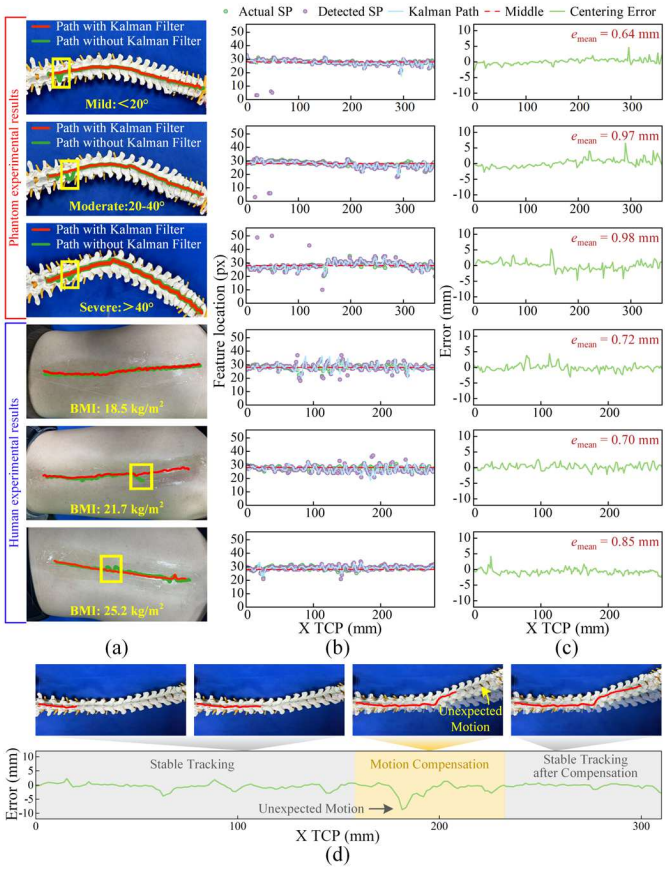


Fig. 4. Results of the feature tracking experiments. (a) The robotic scan trajectories. (b) Spinous process (SP) positions and the corresponding continuous paths. (c) Centering errors during scanning. (d) Adaptation of the system to small-scale lateral movements.

centering error (see Fig. 4(c)) was 0.86 mm for phantom trials and 0.76 mm for human trials. These results demonstrate the high accuracy of our approach in maintaining the probe position. Furthermore, Table III provides a quantitative comparison with other state-of-the-art robotic US systems, highlighting the superior tracking accuracy achieved by our approach. This high level of accuracy allows our system to effectively handle sudden changes in the environment, including unexpected patient movements, as shown in Fig. 4(d). *A comparison video of the scanning process can be found in Movie Part I.*

In addition, we adopted the method described in [47] to fit the detected SP points using a sixth-order polynomial curve. The Cobb angle, which serves as an angular measure to evaluate the degree of spinal curvature, can be calculated by:

$$CobbAngle = \arctan\left(\frac{\lambda_{i-1} - \lambda_i}{1 + \lambda_{i-1}\lambda_i}\right) \tag{15}$$

where λ is the slope between the stationary points with zero partial derivatives of the fitting curve.

Ten independent tests were conducted to compare the values of Cobb angle with ground truth (X-ray images) across various scoliosis types. Statistical results are shown in Table II. The mean relative error (MRE) and root mean square error (RMSE) remain relatively consistent under different conditions, and the overall measurement error falls within the clinically acceptable

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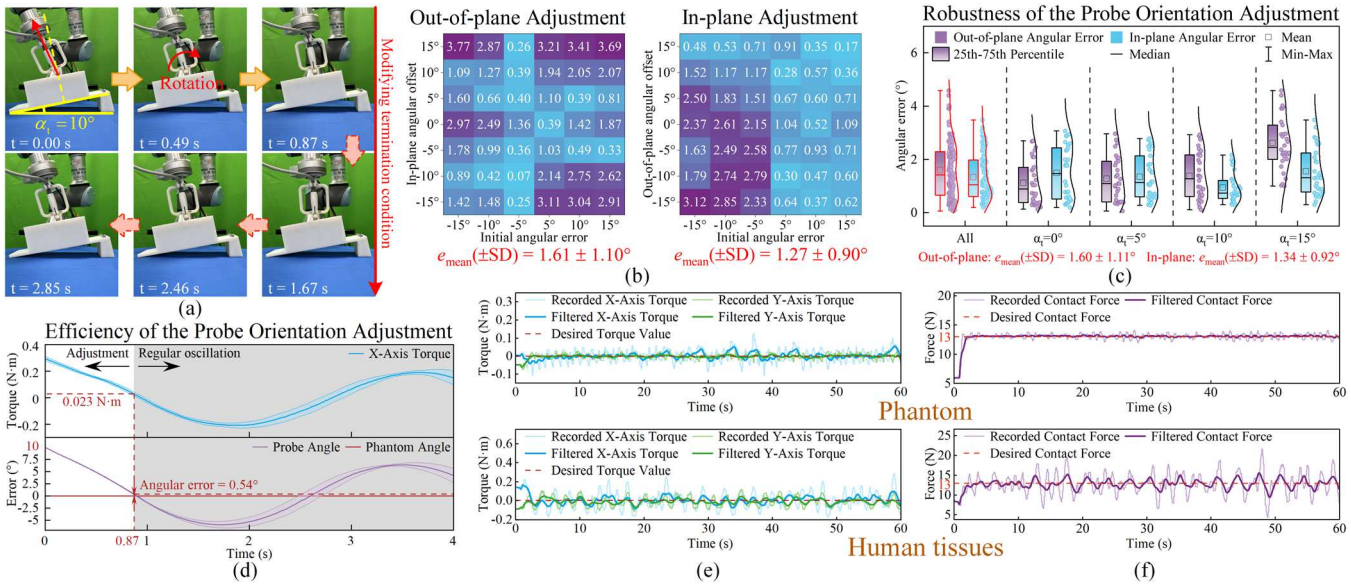


Fig. 5. Results of experiments B and C. (a) Experimental records. (b) Angular errors in accuracy experiments. (c) Adjustment results on various constraint surface. (d) Orientation adjustment efficiency for 10° error. (e)-(f) Variations in torque and contact force during continuous scanning.

TABLE II
MEASUREMENT RESULTS IN PHANTOM EXPERIMENT

Measurement	Mild	Moderate	Severe
Actual angle	16.46 ± 2.02°	26.3 ± 3.15°	48.26 ± 4.11°
Measured angle	15.47 ± 1.89°	25.16 ± 3.66°	47.16 ± 3.94°
MAE±SD	1.02 ± 0.88°	1.31 ± 0.77°	1.2 ± 0.62°
MRE	6.03%	5.15%	2.47%
RMSE	1.32°	1.5°	1.34°
Error range	0.09°-2.39°	0.07°-2.89°	0.44°-2.23°

TABLE III
PERFORMANCE OF DIFFERENT METHODS ON PHANTOM

Methods	Orientation adjustment			Centering error
	out-of-plane	in-plane	time	
Human operator	3.20 ± 1.70°		-	-
Jiang et al. [26]	2.50 ± 1.30°	1.00 ± 0.70°	14.4 s	×
M. Victorova et al. [46]	-	-	-	2.8 ± 1.8 mm
Ning et al. [27]	0.45 ± 2.72°		4 s	1.3 ± 1.0 mm
Wang et al. [39]	3.98 ± 0.13°	1.69 ± 0.83°	-	×
Our method	1.61 ± 1.10°	1.27 ± 0.90°	0.87 s	0.9 ± 0.2 mm

×: method is not appliable. -: value is unknown.

range (5°) [18]. It indicates the robustness of our path planning algorithm in precisely centering features during continuous scanning. This capability facilitates accurate estimation of the Cobb angle, which can assist clinicians in diagnosing spinal disorders, particularly scoliosis.

B. Effectiveness of the Probe Orientation Adjustment

1) Accuracy of the probe orientation adjustment algorithm: The accuracy of the probe orientation adjustment is evaluated based on the angular error between the stable orientation of the probe and the normal direction of the contact surface. To ensure the proposed algorithm is independent of prior knowledge regarding the initial probe orientation, a series of out-of-plane adjustment tests with various initial errors and in-plane angular offsets were implemented. Similarly, the in-plane tests were set. Five experiments were performed for each initial condition, and the mean error was computed as the final result. To simulate the non-planar nature of human back tissues observed in clinical practice, all tests were conducted on a gel phantom placed on a tilted surface, as shown in Fig. 5(a).

The results of the probe orientation adjustment, obtained at a constant contact force of 13 N [19], are listed in Fig. 5(b). The average out-of-plane error (mean ± standard deviation, SD) and in-plane error (mean ± SD) across all tests are 1.61 ± 1.10° and 1.27 ± 0.90°, respectively. The proposed method demonstrates superior accuracy compared to manual scanning, which has been reported to yield an angular error of 3.20±1.70° [39]. It is noteworthy that the in-plane adjustment exhibits a lower overall error margin, possibly due to the mechanical design of the probe. Specifically, the high length-to-width ratio of the linear probe enhances its sensitivity to angle changes during in-plane adjustment, resulting in more precise outcomes. Furthermore, a quantitative comparison of our method with existing state-of-the-art approaches is summarized in Table III. The accuracy of the proposed method is comparable to that of the state-of-the-art approach [26], which reports an out-of-plane error of 2.50 ± 1.30° and an in-plane error of 1.00 ± 0.70°. However, our method is considered more appropriate for clinical applications, since the approach presented in [26] necessitates continuous rotation at a given position, which would affect the efficiency of US

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scanning. *A genuine probe adjustment process can be found in Movie Part II.*

2) Robustness of the probe orientation adjustment algorithm: To investigate the robustness of the proposed probe orientation adjustment algorithm, validations were conducted using a gel phantom placed at various angles (0°, 5°, 10°, 15°). These angles were selected to reflect the typical surface tilts observed in the human back due to physiological curvature and individual variability. For the out-of-plane tests, 5 trials were carried out for different initial in-plane angular offsets (-10°, -5°, 0°, 5°, and 10°) and initial errors (-15°, -10°, -5°, 5°, 10°, 15°). Similar conditions were set for the in-plane tests. The statistical results obtained from 1200 trials are shown in Fig. 5(c). It can be seen that both out-of-plane and in-plane adjustments implemented by our method exhibit higher accuracy compared to manual operations. Specifically, the mean error for the 600 out-of-plane trials is $1.60 \pm 1.11°$, with over 93% of the trials achieving errors below 3.20°. Similarly, for the 600 in-plane trials, the mean error is $1.34 \pm 0.92°$, and over 97% of these trials exhibit errors below 3.20°. Notably, errors exceeding 4° were primarily encountered during out-of-plane adjustments, particularly when a 15° constraint surface and an initial error of 15° were present. This phenomenon can be attributed to potential fluctuations in contact force during large-angle adjustments on tilted surfaces. Overall, the proposed probe orientation adjustment algorithm has demonstrated robust performance across a wide range of constraint surfaces with various tilted angles, effectively satisfying the requirements for unstructured human tissues.

3) Efficiency of the probe orientation adjustment algorithm: Given the surface properties of the human back, a set of out-of-plane adjustment trials with a specific initial error of 10° were executed to assess the efficiency of the proposed method. To analyze probe motion patterns during optimization, the termination condition of the algorithm was modified from the attainment of a predefined threshold to the completion of a fixed timestep. Statistical outcomes from 10 repeated trials are shown in Fig. 5(d). The results demonstrate that the orientation of the probe could be rapidly adjusted to the desired angle, but it subsequently exhibits oscillations that resemble a sine wave rather than remaining stationary. Upon modification of the termination condition, oscillations are expected to occur, as the RL agent is trained to reach a predefined threshold. The mean angle during the oscillation process is defined as the stable angle, with the probe taking an average of 0.87 s to initially converge to the stable angle. It is better than the mean adjustment time of 0.9 s reported in [27]. Furthermore, an additional duration of 3.1 s is requisite for stabilizing the probe, as noted in [27]. A broader comparison of efficiency across different methods is presented in Table III, which further highlights the advantages of our approach. Once the stable angle is reached, the torque correspondingly decreases to 0.023 N·m, signifying that the predefined threshold termination meets the task requirements. Our proposed probe orientation adjustment method indicates satisfactory efficiency, with a stable error of 0.54°, which meets the precision requirements for US imaging applications.

4) Validation of the performance in continuous scanning: Guided by the relationship between torque and angular error in the efficiency experiment, the torque variation was employed to validate the performance of our probe orientation adjustment method in continuous scanning. The desired contact force was set at 13 N, the scanning speed at 4 mm/s, and controller parameters were specified as $\alpha = 0.001$ m/Ns and $\beta = 0.001$. From Fig. 5(e), it can be seen that the actual torque values along the *X*-axis and *Y*-axis closely match the desired values (0 N·m) for both the phantom and human tissues. Particularly, in the case of the phantom, smaller torque errors were observed, with maximum torque errors of 0.05 N·m along the *X*-axis and 0.01 N·m along the *Y*-axis. In the volunteer trials, despite larger variations in actual torque values due to respiratory motion, the mean torque errors along the *X*-axis and the *Y*-axis are only 0.01 ± 0.04 N·m and -0.01 ± 0.03 N·m, respectively. Overall, these results demonstrate that the normal probe orientation can be maintained using the proposed probe orientation adjustment method during continuous scanning. *Several autonomic continuous scanning processes can be found in Movie Part III and IV.*

C. Validation of the Stable Contact Force Control

To further evaluate the proposed control algorithm for stable contact force, two sub-experiments were conducted using the phantom and volunteers. In both trials, identical controller parameters were employed as the same as in continuous scanning trials, with resulting contact force measurements being presented in Fig. 5(f). Initial transient variations in contact force were observed due to the manual placement of the probe. During the stable phase on the phantom, the mean error (±SD) was 0.34 ± 0.25 N. Our method indicated efficacy even under respiratory movements, as evidenced by a force tracking error of 1.01 ± 0.75 N in human tissues. This performance is deemed satisfactory for spinal US imaging, validating the effectiveness of our proposed algorithm across different contact environments and physiological movement interferences.

D. Assessment of Ultrasound Image Quality

1) Repeatability of US image acquisition: The ARUSS is designed to optimize contact with human tissues, aiming for the acquisition of high-quality US images with good repeatability. In this test, as shown in Fig. 6(a), the US probe was maneuvered to engage three predefined positions on the test object, adopting three initial orientations. Subsequently, the structural similarity (SSIM) index [48] was used to analyze the consistency among the stable US images. Each test was repeated five times on both a phantom and a human volunteer. The statistical analysis of SSIM values across all trials is presented in Fig. 6(b). In both the phantom and human tissue scenarios, the proposed ARUSS demonstrated superior performance, achieving higher mean SSIM values (Phantom: 0.91; Human: 0.36) compared to manual scanning (Phantom: 0.85; Human: 0.32). To further illustrate the consistency of the acquired images, Fig. 6(c) and (d) display representative US images acquired from identical positions in phantom and

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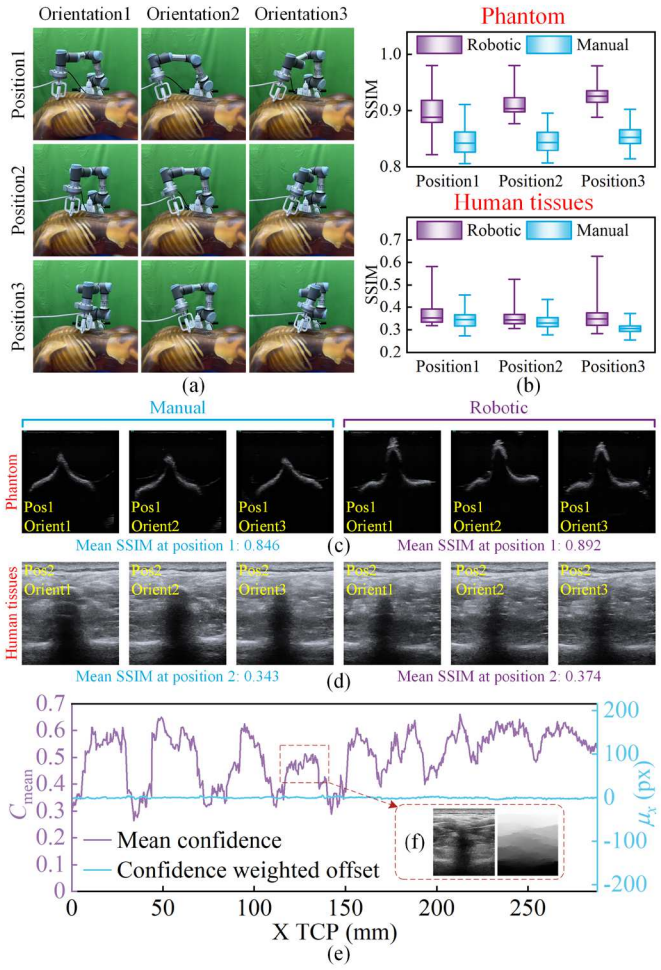


Fig. 6. Results of experiment D. (a) Experimental setup for the ARUSS repeatability. (b) SSIM results of US images. (c) Typical US images and overall SSIM from phantom. (d) Typical US images and overall SSIM from human tissues. (e) Mean confidence values and the offsets of confidence map weighted barycenter during scanning. (f) Typical US image and its corresponding confidence map.

human tissues, respectively, using both manual and robotic scanning. These qualitative examples complement the statistical analysis, visually underscoring the enhanced repeatability achieved by ARUSS. These results indicate that the proposed ARUSS has an outstanding ability to accurately detect nuanced variations in orientation and position, which has proved to be challenging for human operators. *A comparison video can be found in Movie Part V.*

2) US image quality assessment for continuous scanning: The objective of this study is to obtain high-quality US images of the spine. To validate this, we used the confidence map algorithm [49] to assess the sequential US images procured from human tissues. Evaluation metrics included the mean confidence value C_{mean} and the offset μ_x of the weighted barycenter, measured along the X-axis of the image frame $\{I\}$.

$$C_{mean} = \frac{1}{H \cdot W} \sum_{(h,w)\in\mathbf{\Omega}} C(h,w) \tag{16}$$

$$\mu_x = \frac{1}{C_{\mathbf{\Omega}}} \sum_{(h,w)\in\mathbf{\Omega}} w \cdot C(h,w) \tag{17}$$

where $\mathbf{\Omega} = [1 \cdots H] \times [1 \cdots W]$ and $C_{\mathbf{\Omega}} = \sum C(h,w)$ denote the collection of US image pixels and the total confidence, respectively. Fig. 6(f) shows a typical spinal US image and its corresponding confidence map. Fig. 6(e) documents the variations in the mean confidence value and the X-axis offset of the weighted barycenter during the scanning process. The average ($\pm$SD) of the C_{mean} is 0.51 $\pm$ 0.09, which concurs with our anticipations. Notably, bone tissues exhibit lower mean confidence values than general soft tissues, since their higher acoustic impedance causes the inherent shading in confidence maps. In addition, the observed cyclic variation in mean confidence aligns with the physiological chain of the spine. The mean ($\pm$SD) of the μ_x consistently approximates 0 (1.24 $\pm$ 0.99 pixels), indicating an appropriate probe orientation. These results demonstrate the reliability of the proposed ARUSS in acquiring high-quality spinal US images. *All experiments were conducted under the safety mechanism demonstrated in Movie Part VI.*

VII. Conclusion

In this study, we introduced an end-to-end ARUSS based on dual agents for the spine in external-vision-independent environments. The system allows for precise adjustment of the probe's position and orientation, enabling the acquisition of high-quality US images. To address the unique challenges posed by the chain-like structure of the spine, a multi-task network and a feature tracking algorithm based on real-time US images were proposed. On this basis, a learning-based approach was presented to establish the nonlinear relationship between contact force and relative probe orientation. This framework was integrated with an F2V algorithm for multi-objective optimization, maintaining the normal probe orientation and stable contact force. Experimental results on phantoms and volunteers demonstrated the system's autonomous capability in performing spinal US scanning and accurately tracking spinal features. The system provides robust performance in optimizing probe orientation and contact force, facilitating repeated acquisition of high-quality US images.

However, there are still some limitations to our study. Firstly, our system is not yet fully autonomous, as it necessitates manual intervention to establish the initial probe position, although this procedure can be concurrently executed with the application of the US gel. Secondly, the proposed control framework only uses the contact force as the observation, whereas the potential information of US images can also benefit for enhanced scanning control. In the future, we plan to utilize both the force and image information to improve the quality of robotic image acquisition.

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